

Mounika Geriki | AI-Software Engineer

TX, USA | +1 (934) 949-8567 | germounika3@gmail.com | [LinkedIn](#)

SUMMARY

AI Software Engineer with 4+ years of experience building AI-powered applications, machine learning solutions, and scalable backend systems using Python, SQL, and cloud technologies. Experienced in developing Agentic AI, Retrieval-Augmented Generation (RAG), and LLM-based applications, integrating enterprise systems through REST APIs, and deploying production-ready AI services with Docker, Kubernetes, and CI/CD pipelines. Skilled in NLP, MLOps, vector databases, and cloud-based AI solutions across AWS, Azure, and Google Cloud Platform.

SKILLS

Programming Languages: Python, SQL

AI & Generative AI: Large Language Models (LLMs), Agentic AI, AI Agents, Retrieval-Augmented Generation (RAG), LangChain, OpenAI API, Prompt Engineering, Tool Calling, Model Context Protocol (MCP), A2A Protocol, AGNTCY, Knowledge Graphs, Vector Databases, FAISS, Semantic Retrieval

Machine Learning & NLP: Machine Learning, Natural Language Processing (NLP), Feature Engineering, Model Deployment, Model Versioning, Model Drift Monitoring, MLflow, spaCy, Tesseract OCR, Scikit-learn

Backend & Software Engineering: Microservices, Spring Boot, REST APIs, Webhooks, Flask, Multi-Tenant SaaS Applications, Event-Driven Architecture, API Integration

Data Engineering: ETL Pipelines, Apache Spark, Data Processing, SQL Query Optimization, Database Schema Design, MySQL, PostgreSQL

Cloud & MLOps: AWS, Azure, Google Cloud Platform (GCP), Docker, Kubernetes, Azure Kubernetes Service (AKS), Amazon Elastic Kubernetes Service (EKS), CI/CD Pipelines

Enterprise AI & Integrations: Salesforce Integration, ServiceNow Integration, Jira Integration, Enterprise AI Systems, AI Workflow Automation

AI Security & Observability: LLM Guardrails, PII Redaction, Responsible AI, TRAPS Governance, OpenTelemetry, Logging Frameworks, Observability Pipelines

Developer Tools & Platforms: Git, Streamlit

Development Methodologies: Agile Scrum, Code Review, Debugging, Performance Optimization

EXPERIENCE

AI-Software Engineer Aisera USA (Remote)	Jul 2025 – Present
- Architected Python-based microservices for the Aisera Unify platform, enabling real-time orchestration of autonomous AI agents through open communication frameworks including A2A, MCP, and AGNTCY. - Developed RAG pipelines, vector databases, and knowledge graph integrations to retrieve enterprise context, improving AI response relevance and reducing knowledge retrieval latency by 28%. - Integrated REST APIs, webhooks, and Salesforce, ServiceNow, and Jira connectors to enable secure communication between AI agents and enterprise systems, accelerating automated task execution by 35%. - Implemented TRAPS governance, PII redaction, and LLM guardrails while evaluating AI responses for answer relevance, hallucination mitigation, response latency, and tool-call reliability to support secure and dependable enterprise AI interactions. - Engineered Docker, Kubernetes, and event-driven services that connected conversational AI with enterprise automation workflows, reducing end-to-end request processing time by 24% through low-latency orchestration. - Built OpenTelemetry logging and observability pipelines to monitor LLM token usage, agent execution, response latency, and session memory, improving production visibility across multi-agent systems.	
ML Engineer Mphasis India	Sep 2021 – Dec 2023
- Engineered Python, spaCy, Tesseract OCR, and NLP pipelines to extract structured information from unstructured enterprise documents, enabling automated processing for banking and insurance workflows with 95% extraction accuracy. - Implemented PACE-ML, MLflow, Azure/AWS, and CI/CD pipelines to operationalize machine learning models, automating model versioning, deployment, and production monitoring while reducing deployment time by 40%. - Designed Apache Spark and SQL feature engineering and ETL pipelines to process large-scale enterprise datasets, improving data preparation efficiency by 30% for downstream machine learning workloads. - Containerized Docker-based machine learning services and orchestrated deployments on Kubernetes (AKS/EKS) to deliver scalable, real-time inference APIs for enterprise AI applications. - Integrated MLflow model lifecycle management with drift monitoring and validation workflows to support reliable production performance and continuous model governance across cloud environments. - Collaborated with cross-functional engineering teams to develop Generative AI and LLM proof-of-concepts during the early Mphasis.ai initiative, supporting enterprise AI solution development and cognitive automation use cases.	
Software Developer Freshworks India	Sep 2020 – Aug 2021
- Developed Spring Boot backend features for multi-tenant SaaS applications, delivering scalable and maintainable functionality that supported enterprise customer workflows. - Designed REST APIs and integrated internal services with third-party applications, enabling secure data exchange and improving system interoperability across product modules. - Optimized MySQL/PostgreSQL queries and updated database schemas to support secure multi-tenant data management, reducing query execution time by 22% through efficient indexing and query tuning. - Collaborated in Agile Scrum ceremonies, performed peer code reviews, resolved production defects, and contributed to CI/CD deployment pipelines to ensure reliable feature releases. - Improved application performance using profiling and debugging tools, identifying latency bottlenecks that increased API response efficiency by 18% and enhanced overall application responsiveness. - Maintained Git-based version control practices while partnering with cross-functional engineering teams to deliver high-quality backend enhancements for globally deployed SaaS products.	

PROJECTS

InsightPilot – Agentic SQL & Analytics Platform | Python, SQL, LangChain, FAISS, Streamlit, OpenAI API

- Engineered an agentic AI analytics platform using LangChain and the OpenAI API to translate natural language business questions into validated SQL queries and generate executive-ready insights from structured data.
- Implemented SQL validation guardrails to detect and block unsafe write/delete operations, ensuring secure query execution while improving reliability and governance for business users.
- Integrated FAISS for semantic retrieval and built an interactive Streamlit interface, reducing query response time by 32% and delivering a more efficient analytics experience.

TravelWise – Agentic RAG Travel Assistant | Python, Flask, LangChain, FAISS, GCP, REST APIs

- Developed an agentic Retrieval-Augmented Generation (RAG) travel assistant using LangChain, FAISS, and Flask to deliver context-aware travel recommendations through semantic search and tool-calling workflows.
- Designed REST API integrations and retrieval pipelines to enrich LLM responses with location-specific information, improving answer relevance and reducing hallucinations through retrieval-based grounding.
- Evaluated the application on 50 benchmark queries, achieving 91.4% answer accuracy, a 2.4% error rate, and an average response time of 2.8 seconds.

CERTIFICATIONS

- Completed 5-Day Gen AI Intensive Course — Google x Kaggle
- AI Agent Fundamentals — Databricks
- Generative AI in Data Analytics — Meta
- Google Cloud IAM and Networking for AWS Professionals — Google Cloud

EDUCATION

Master of Science in Data Science

Stony Brook University | Stony Brook, NY, USA

Dec 2025

Bachelor of Engineering in Electronics and Communication Engineering

Bangalore Institute of Technology | Bangalore, India

Aug 2021